

JEEVAN JAYASURIYA

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SUMMARY

Ph.D. student in Industrial and Systems Engineering focused on neuroergonomics, human factors, machine learning, deep learning, mechatronics, and biomechanics, with strong expertise in multimodal data analysis. Hands-on experience through research and prior industrial work in user-centered system design. Passionate about applying neuroergonomics and engineering principles toward effective human-machine interface design and process optimization.

EDUCATION

UNIVERSITY OF WISCONSIN - MADISON

Ph.D., Industrial & Systems Engineering (Expected May 2027)

CLARKSON UNIVERSITY, POTSDAM, NEW YORK

M.S., Mechanical Engineering, 2022 - 2023

THE OPEN UNIVERSITY OF SRI LANKA, NUGEGODA

B.Tech., (Hons.), Mechatronics Engineering, 2013 - 2017

UNIVERSITY OF SRI JAYEWARDENEPURA, SRI LANKA

B.S., (Hons.), Physics, 2011 - 2015

EXPERIENCE

UNIVERSITY OF WISCONSIN - MADISON

Madison, Wisconsin

Graduate Research Assistant - NeuroErgonomics Lab 2024-Present

- Collaborating with the Neuroscience Laboratory, NASA Johnson Space Center (Houston, TX) to characterize and mitigate the combined effects of altered gravity and fatigue from cognitive workload and sleep deprivation on astronaut sensorimotor performance.
- Designed and executed complex, long-duration human subject experiments, integrating synchronized multimodal physiological data collection using the Lab Streaming Layer (LSL).
- Performed data acquisition, preprocessing, analysis, and time-series modeling of multimodal physiological data to uncover complex human factors underlying sensorimotor performance, using ambulatory brain imaging (fNIRS), eye tracking (Tobii Glasses 3), surface EMG (Noraxon), full-body kinematics (Xsens), heart rate monitoring (Actiheart; Empatica Embrace Plus), and force plate measurements (AMTI).

CLARKSON UNIVERSITY

Potsdam, New York

Graduate Research Assistant - Astronautics and Robotics Lab 2022-2023

- Applied movement kinematics-driven musculoskeletal simulations to guide adaptive exoskeleton design for firefighter tasks, integrating biomechanics with mechatronics and mechanical engineering concepts.
- Gained hands-on experience in CAD design (SolidWorks), feedback control, microcontrollers, actuator and servo systems, sensor fusion, and system-level integration, along with prototyping, material selection, and 3D printing.

UNIVERSITY OF SRI JAYEWARDENEPURA

Gangodawila, Sri Lanka

Lecturer – Embedded Systems 2020-2021

- Lectured undergraduate courses in Control Systems, Mobile App Development, Advanced Programmable Logic Controllers (PLCs), Object-Oriented Programming (OOP) for Embedded Systems, and Data Communication.

PEOPLES BANK Colombo, Sri Lanka
Assistant Manager – Corporate Banking 2017-2020

- Conducted feasibility studies for credit package approvals in the Corporate Banking Division, handling large credit profiles from leading private and government institutions in Sri Lanka, while applying analytical and project management skills.

SRI LANKA ATOMIC ENERGY BOARD Wellampitiya, Sri Lanka
Scientific Officer 2017

- Led the Radiation Protection Section, managing a team to ensure compliance with international radiation safety standards and regulations for radiation workers in Sri Lanka.

ENGINEERING TRAINEE ROLES Sri Lanka
Trainee 2015-2017

- Arthur C. Clarke Institute for Modern Technologies – Contributed to research and development projects in robotics and automation.
- OREL Corporation – Designed mechanical, pneumatic, and electro-pneumatic automation circuits using SolidWorks and FluidSim.

PUBLICATIONS

- **Jayasuriya, J.**, Bazzocchi, M. C. F., Fite, K., & Martinez, M. (2026). Firefighter motion capture data set and biomechanical analysis of task-specific gear effects and loading asymmetries. *Ergonomics*
<https://doi.org/10.1080/00140139.2025.2605062>
- **Jayasuriya, J.**, Bazzocchi, M. C. F., Fite, K., & Martinez, M. (2026). Modeling and analysis of firefighter musculoskeletal biomechanics for lower-extremity exoskeletons. *Journal of Safety Research*
<https://doi.org/10.1016/j.jsr.2026.04.001>
- Shanghavi, A. A., **Jayasuriya, J.**, Cai, Y., & Mehta, R. K. (2025). Pulse Rate Variability From Wearable Wristwatches as a Surrogate for Heart Rate Variability? We Think Not Yet. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, <https://doi.org/10.1177/10711813251357929>
- Hayes, J., Fairchild, B., Nichols, K., **Jayasuriya, J.**, & Mehta, R.K. (2024). Selective Sensorimotor Impairments With Mental Fatigue Associated With G-Transitions During and After Spaceflight. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, <https://doi.org/10.1177/10711813241275073>
- **Jayasuriya, J.**, Moser, I., & de Mel, R. (2022). Automated Water Dispensing System for Controlling Fires in Coal Yards. *International Journal of Coal Science and Technology*, <https://doi.org/10.1007/s40789-022-00488-y>

TECHNICAL SKILLS

Physiological & Biomechanical Data Analysis:

Functional near-infrared spectroscopy (fNIRS), Electrophysiological signals (EMG: Noraxon; ECG/heart rate: Actiheart, Empatica Embrace Plus), Eye tracking (Tobii Glasses 3, Tobii Pro Lab), Force plate data (AMTI), Full-body kinematics

Machine Learning, Statistics & Data Visualization:

Python, PyTorch, TensorFlow, C++, R, MS Office Suite (Word, Excel, PowerPoint)

Engineering & Simulation Tools:

MATLAB, Simulink, SolidWorks, PTC Creo, Autodesk 3ds Max, FluidSim, Visual Studio, Arduino, OpenCV, Java

AWARDS & AFFILIATIONS

Best Student Work Award in the Aerospace Systems Technical Group (ASTG) in ASPIRE (2024 & 2025)

Academic Excellence Awards in Engineering (2016 & 2017)

Associate Member, Institute of Engineers Sri Lanka (IESL) (2018 - present)

Student Affiliate Human Factors and Ergonomics Society (HFES) (2024 - present)